Assignment -7

(Series & Patterns using Loops)

1. Write a program in Java to print and display the sum of the series

   $$1+ 1/2 + 1/3 + 1/4 + 1/5 + .. 1/N.$$

2. Write a program in Java to print and display the sum of the series

   $$1 + 1/x + 1/x^2 + 1/x^3+.....+1/x^n.$$

3. Write a program in Java to print and display the sum of the series

   $$1 + 1/2! + 1/3!+..... +1/n!$$

4. Write a program in Java to print and display the sum of the series

   $$1 + 2^3 + 3^2 + 4^3 + ...+n$$

5. Find all the prime numbers in the range from m to n whose sum of digits is also a prime number.

6. Generate the following patterns:

   ```
   1 0 0 0 1
   0 1 0 1 0
   0 0 1 0 0
   0 1 0 1 0
   1 0 0 0 1
   ```

7. Generate the following patterns:

   ```
   5 4 3 2 1
   5 4 3 2
   5 4 3
   5 4
   5
   ```